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Proposed simulation study: Hausman–Taylor under weak internal instruments

We propose to study the finite-sample performance of the Hausman–Taylor estimator in a static panel data model with both time-varying and time-invariant regressors. The model is given by

$$y_{it} = \beta_u x_{u,it} + \beta_c x_{c,it} + \gamma_u z_{u,i} + \gamma_c z_{c,i} + c_i + u_{it}.$$

Here, $x_{u,it}$ is time-varying and exogenous, $x_{c,it}$ is time-varying and correlated with c_i, $z_{u,i}$ is time-invariant and exogenous, and $z_{c,i}$ is time-invariant and correlated with c_i.

We simulate balanced short panels, for example with $N = 500$, $T = 10$, and repeat the simulation over many Monte Carlo replications. The key part of the data-generating process is

$$z_{c,i} = \pi \bar{x} * u, i + \rho_z c_i + r_i, \qquad \bar{x} * u, i = \frac{1}{T} \sum_{t=1}^{T} x_{u,it}.$$

The parameter ρ_z controls the endogeneity of $z_{c,i}$, while π controls the relevance of the Hausman–Taylor instrument $\bar{x}_{u,i}$. We vary

$$\pi \in [0, 1]$$

to compare strong and weak internal instruments. Optionally, we also consider a scenario in which $x_{u,it}$ is incorrectly treated as exogenous, making the instrument invalid.

In the main simulation design, we keep ρ_z fixed at a positive value, for example $\rho_z = 0.7$, so that $z_{c,i}$ is clearly endogenous.

We compare POLS, RE, FE, CRE, and Hausman–Taylor. POLS and RE are included as benchmarks, but are expected to be biased when regressors are correlated with c_i. FE should estimate the time-varying coefficients consistently, but cannot estimate the time-invariant effects. Hausman–Taylor is the main estimator of interest, since it can estimate the endogenous time-invariant effect γ_c if the internal instrument is valid and relevant. Estimator performance is evaluated using bias and RMSE.